Connecting Devices & Spanning Tree Protocol (STP)

Coverage: Repeaters, Hubs (L1), Bridges & Layer 2 Switches (MAC Learning), Spanning Tree Protocol (IEEE 802.1D STP), Routers

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1. Connecting Devices Taxonomy

- Hubs (L1): Single collision/broadcast domain. Switches (L2): Microsegmented collision domains. Routers (L3): Separate broadcast domains.

2. Spanning Tree Protocol (STP)

- Breaks switching loops: 1. Root Bridge election (lowest Bridge ID), 2. Root Ports, 3. Designated Ports, 4. Blocking ports.

■ KTU University High-Yield Exam Questions

- Q: Compare Hubs, Switches, and Routers and explain STP algorithm. (8 Marks)